JR Validated Environment — SPC Module

User Guide for Statistical Process Control

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1. How to Use This Guide

This guide describes the five SPC control chart scripts available in the JR Validated Environment. Each script produces a different chart type suited to a specific data structure and monitoring goal. Choose the chart that matches your process data:

- One measurement per time point (individual units, batches, shifts): I-MR chart
- Small subgroups of measurements per time point, subgroup size 2–10: X-bar/R chart
- Subgroups of any size, or subgroup size > 10: X-bar/S chart
- Count of nonconforming units (pass/fail) per inspection lot: P-chart
- Count of defects per unit, constant inspection unit size: C-chart

All scripts apply the 8 Western Electric (WE) rules and output an IN CONTROL / OUT OF CONTROL verdict. The continuous measurement scripts (I-MR, X-bar/R, X-bar/S) support optional user-specified historical control limits. See Section 3 for interpretation guidance and Section 6 for the full WE rule definitions.

2. Quick Reference Table

Find the row that matches your data, then go to the section listed.

My data looks like…	Script	Section
I collect one measurement per time point (one batch, one shift, one unit per day)	jrc_spc_imr	4.1
I collect small groups of units at each time point, subgroup size 2–10	jrc_spc_xbar_r	4.2
I collect subgroups at each time point, subgroup size > 10 or any size	jrc_spc_xbar_s	4.3
I record how many units fail inspection in each batch or lot (pass/fail count)	jrc_spc_p	4.4
I count individual defects per unit (one unit can have multiple defects)	jrc_spc_c	4.5

3. Understanding the Results

3.1 Control Limits

Control limits (UCL and LCL) are calculated from the data itself — they are not specification limits. They represent the expected range of variation for a stable process (approximately ±3σ from the centreline). A point outside the control limits, or a non-random pattern within them, is a signal that something has changed in the process.

INFO: IN CONTROL does not mean the process meets specification. A process can be perfectly stable and predictable while still producing out-of-spec results. Whether values meet specifications is answered by capability analysis (jrc_capability), not by a control chart.

3.2 IN CONTROL / OUT OF CONTROL Verdict

The terminal output ends with one of two verdicts:

- IN CONTROL — no WE rule violations detected. The process is stable and predictable.
- OUT OF CONTROL — one or more WE rule violations detected. The script lists the rule numbers and the affected point IDs. Investigate for assignable (special) causes before using the process outputs for product release.

3.3 Western Electric Rules

All eight WE rules are applied to the primary chart (I chart, X-bar chart, P chart, C chart). The secondary chart (MR chart, Range chart, S chart) has only Rule 1 applied. Section 6 defines all eight rules. Key guidance:

- Rule 1 (point beyond 3σ): highest priority signal — investigate immediately.
- Rules 2, 3, 5 (runs and trends): indicate a sustained process shift or drift.
- Rules 4, 6, 7, 8 (patterns): indicate cycles, over-control, or stratification.

3.4 Historical Limits

The I-MR, X-bar/R, and X-bar/S scripts accept optional --ucl and --lcl arguments. Use historical limits when you want to compare current production against a previously established baseline rather than computing limits from the current data. Historical limits are displayed on the chart alongside the data but the verdict and WE rules are still evaluated against these user-supplied values.

3.5 Variable Control Limits (P-chart)

The P-chart computes per-subgroup control limits when subgroup sizes (n) vary between lots. Larger subgroups produce narrower limits; smaller subgroups produce wider limits. The chart displays limit lines that step with each subgroup. WE pattern rules are applied using standardised z-values to avoid false signals from the changing limit width.

4. Per-Script Reference

4.1 jrc_spc_imr — Individuals and Moving Range (I-MR)

When do I use this?

Use when you have one measurement per time point — one reading per shift, one batch result per day, one patient measurement per visit. The I chart monitors the process level and the MR chart monitors process variation (as the absolute difference between consecutive observations). This is the right chart when subgrouping is not practical.

What do I need?

Argument	What to provide
data.csv	CSV with columns: id, value. The 'id' column labels each observation (shift number, date, batch ID). The 'value' column is the numeric measurement. Minimum: 3 observations (to compute at least 2 moving ranges).
--ucl <value>	Optional. User-specified Upper Control Limit for the I chart. Use when applying historical control limits established from a prior stable period.
--lcl <value>	Optional. User-specified Lower Control Limit for the I chart.

Example

```
jrc_spc_imr batch_results.csv
jrc_spc_imr daily_weight.csv --ucl 10.75 --lcl 9.25
```

What to look for in the output

Check the terminal verdict. If OUT OF CONTROL, the listed rule numbers and point IDs tell you where to investigate. The two-panel PNG shows the I chart on top (with 1σ and 2σ zone lines for WE rules) and the MR chart below. Out-of-control points are shown in red. A signal on the MR chart indicates a sudden change in variability — investigate before interpreting the I chart.

4.2 jrc_spc_xbar_r — X-bar and Range

When do I use this?

Use when you measure a small group (subgroup) of units at regular intervals — for example, five parts pulled from a line every hour, or three readings at the start of each shift. The X-bar chart tracks the subgroup mean; the Range chart tracks within-subgroup spread. Best suited for subgroup sizes of 2–10. For larger subgroups, use jrc_spc_xbar_s.

What do I need?

Argument	What to provide
data.csv	CSV with columns: subgroup, value (long format — one row per observation). The 'subgroup' label groups observations collected at the same time point. All subgroups must have the same number of observations (balanced design). Minimum: 2 subgroups, subgroup size n ≥ 2.
--ucl <value>	Optional. User-specified UCL for the X-bar chart.
--lcl <value>	Optional. User-specified LCL for the X-bar chart.

Example

```
jrc_spc_xbar_r hourly_samples.csv
jrc_spc_xbar_r fill_weights.csv --ucl 105.2 --lcl 94.8
```

What to look for in the output

The terminal output shows chart constants (d2, D3, D4, A2) for the detected subgroup size, grand mean, R-bar, control limits, and any WE rule violations. The two-panel PNG shows the X-bar chart on top and the Range chart below. Investigate any Range chart signals first — unstable variation makes the X-bar limits unreliable.

4.3 jrc_spc_xbar_s — X-bar and Standard Deviation

When do I use this?

Use for subgrouped data when subgroup size is larger than 10, or when you want the statistically more efficient standard deviation in place of the range. The X-bar/S chart is valid for any subgroup size ≥ 2. Chart constants (c4, A3, B3, B4) are computed analytically from the gamma function, so no lookup table is required.

What do I need?

Argument	What to provide
data.csv	CSV with columns: subgroup, value (long format — one row per observation). All subgroups must have the same number of observations (balanced design). Minimum: 2 subgroups, subgroup size n ≥ 2.
--ucl <value>	Optional. User-specified UCL for the X-bar chart.
--lcl <value>	Optional. User-specified LCL for the X-bar chart.

Example

```
jrc_spc_xbar_s subgroup_data.csv
jrc_spc_xbar_s tensile_strength.csv --ucl 52.0 --lcl 44.0
```

What to look for in the output

The terminal output shows chart constants (c4, A3, B3, B4), grand mean, S-bar, control limits, and any WE rule violations. The two-panel PNG shows the X-bar chart on top and the S chart below. Investigate S chart signals before interpreting the X-bar chart.

TIP: When subgroup size is between 2 and 10, jrc_spc_xbar_r and jrc_spc_xbar_s will give virtually identical results in practice. Choose R for simplicity; choose S if your organisation's SOP specifies the standard deviation chart or if subgroup size exceeds 10.

4.4 jrc_spc_p — P-chart (Proportion Nonconforming)

When do I use this?

Use when each data point represents an inspection lot where you record the number of units inspected and the number found nonconforming (pass/fail). Each unit is classified as either conforming or nonconforming. Supports variable lot sizes — control limits automatically adjust for each subgroup's sample size.

What do I need?

Argument	What to provide
data.csv	CSV with columns: subgroup, n, defectives. 'subgroup' is a label for each lot or time period. 'n' is the number of units inspected. 'defectives' is the count of nonconforming units (must be ≤ n). Minimum: 2 subgroups. All n ≥ 1. All defectives ≥ 0.

Example

```
jrc_spc_p weekly_inspection.csv
jrc_spc_p final_test_results.csv
```

What to look for in the output

The terminal shows p-bar (weighted overall proportion defective), per-subgroup limits if n varies, and any WE rule violations. The PNG shows each subgroup's fraction defective with stepped per-point limit lines. OUT OF CONTROL signals indicate the defect rate has shifted. IN CONTROL at a high defect rate still requires process improvement — the chart confirms stability, not acceptability.

4.5 jrc_spc_c — C-chart (Count of Defects per Unit)

When do I use this?

Use when you count the number of individual defects on each inspection unit — for example, weld spatter points per panel, cosmetic blemishes per device, solder defects per circuit board, or documentation errors per batch record. A single unit can have multiple defects. The inspection unit must be of constant size or opportunity for defects. The statistical basis is the Poisson distribution.

What do I need?

Argument	What to provide
data.csv	CSV with columns: (subgroup or id), defects. The first column may be named 'subgroup' or 'id' — both are accepted. The 'defects' column must contain non-negative integers. Minimum: 2 subgroups.

Example

```
jrc_spc_c cosmetic_inspection.csv
jrc_spc_c weld_defects.csv
```

What to look for in the output

The terminal shows c-bar (mean defects per unit), sigma (square root of c-bar), UCL, LCL, and any WE rule violations. The PNG shows defect counts per unit with UCL/LCL and 1σ/2σ zone lines. A Rule 1 signal (sudden high count) requires immediate investigation. Pattern rules (runs, trends) suggest a gradual drift in the defect-generating process.

NOTE: The C-chart requires constant inspection opportunity. If your inspection unit varies in size (different lengths of weld, different numbers of solder joints per board), the control limits will be incorrect. In that case, calculate defects per unit of opportunity and use an appropriate U-chart — not currently available as a JR script.

5. Preparing Your Data

Each script expects a CSV file with specific column names. The table below summarises the required columns for each script.

Script	Required columns	Optional columns
jrc_spc_imr	id, value	none
jrc_spc_xbar_r	subgroup, value (long format)	none
jrc_spc_xbar_s	subgroup, value (long format)	none
jrc_spc_p	subgroup, n, defectives	none

jrc_spc_c	subgroup (or id), defects	none

Column names are case-insensitive. Extra columns are ignored. The 'id' and 'subgroup' columns may contain text labels or numbers. Numeric value columns must be numeric. Minimum row counts are enforced; the script will exit with a clear error if the data does not meet the minimum requirements.

INFO: The X-bar/R and X-bar/S scripts use long format: one row per observation, not one row per subgroup. If your data is in wide format (one row per subgroup with multiple value columns), you must reshape it before running the script. Use jrc_convert_csv or standard spreadsheet operations to pivot the data.

6. Western Electric Rules Reference

All scripts apply all 8 Western Electric rules to the primary chart. The secondary variability chart (MR, R, or S) has only Rule 1 applied. The P-chart applies rules using standardised z-values to account for variable control limit widths.

Rule	Signal definition	Likely cause
1	Any single point beyond 3σ (outside control limits)	Immediate special cause — highest priority signal
2	9 consecutive points on the same side of the centreline	Sustained process shift — investigate for a step change
3	6 consecutive points steadily increasing or decreasing	Process drift or trend — investigate for gradual change (tool wear, temperature)
4	14 consecutive points alternating up and down	Over-control or systematic oscillation — investigate for two alternating streams
5	2 of 3 consecutive points beyond 2σ on the same side	Incipient shift — early warning of a process change
6	4 of 5 consecutive points beyond 1σ on the same side	Sustained off-centre — process mean has shifted slightly
7	15 consecutive points within 1σ of the centreline (stratification)	Two or more streams mixed in the data, or over-tight sampling
8	8 consecutive points beyond 1σ on either side (mixture)	Two distinct process streams contributing to the data

When a rule fires, the terminal output lists the rule number and the IDs of the points involved. Use this information to identify the time window to investigate. Begin with Rule 1 signals; pattern rules (2–8) indicate sustained process changes that may not produce individual extreme values.

IMPORTANT: Out-of-control signals require investigation and documented root cause analysis before accepting the process output for product release in a regulated environment. Do not re-run the chart with different data to make signals disappear — address the assignable cause.